EX.NO: 6
5/2/2026

VERIFICATION OF MAXIMUM POWER TRANSFER THEOREM

AIM:

To measure the power absorbed in a load and to verify that the power absorbed in a load is maximum only when load resistance is equal to the source resistance.

APPARATUS REQUIRED:

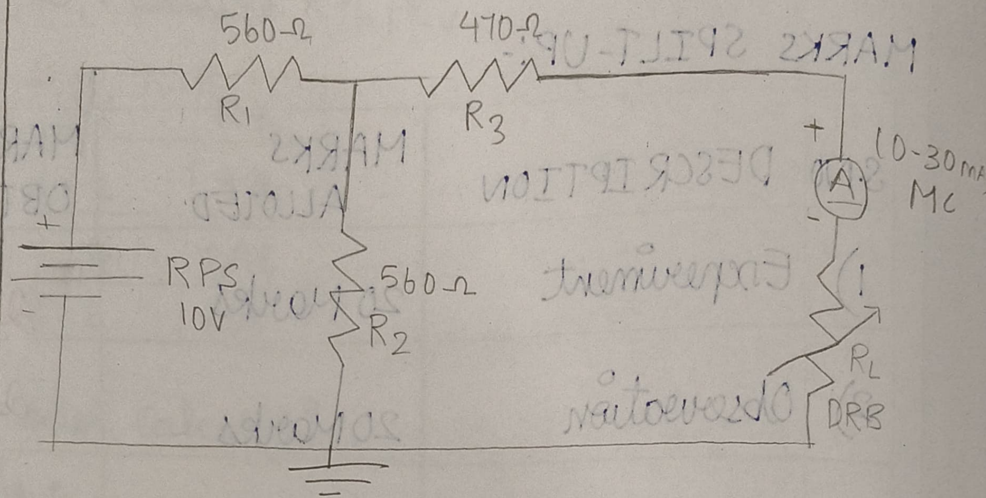
S.NO	NAME OF THE APPARATUS	RANGE/RATING	QUANTITY
1)	Voltmeter	(0-15V) MC	1
2)	Ammeter	(0-500mA) MC	1
3)	Resistors	560Ω 470Ω	2 1
4)	RPS (DC Supply)	15V	1

PROCEDURE:

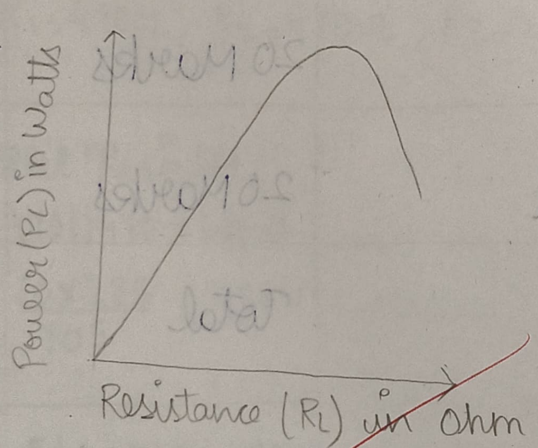
1) Make connections as per the circuit diagram.

2) Change the Resistors R_L whose value close to R_{Th}, Measure the corresponding

CIRCUIT DIAGRAM:



MODEL GRAPH:



V_L, I_L and Calculate the power P_L and enter into the table (2).

3) Plot a graph between R_L and P_L and find the R_L corresponding to maximum power transfer.

4) Verify the measured values of R_L at maximum power transfer as same as calculated and found graphically.

TABULATION:

TABLE 1 : FOR THEORETICAL CALCULATION : R_{Th} =

S.NO	Load Resistance R_L in ohm	Load Current (I_L) in mA	Load Voltage (V_L) in Volts	Load Power (P_L) in Watts
1	500	4.00	2.00	0.0080
2	750	3.33	2.50	0.00833
3	1000	2.86	2.86	0.00816
4	1500	2.00	3.00	0.00600
5	2000	1.67	3.333	0.00556

TABLE (2) : FOR PRATICAL CALCULATION

S.No	Load Resistance R_L in ohm	Load Current (I_L) in amps	Load Voltage (V_L) in Volts	Load Power (P_L) in Watts
1	500	3.90	1.95	0.0076
2	750	3.25	2.44	0.00793
3	1000	2.80	2.80	0.00784
4	1500	1.95	2.93	0.00571
5	2000	1.60	3.20	0.00512

verified
8/10
24/2/26

MODEL CALCULATION:

Given: $R_1 = 560\Omega$; $R_2 = 560\Omega$, $R_3 = 470\Omega$

Supply Voltage = 10 V.

R_{Th}:

$$R_{th} = R_3 + \left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

$$= 470 + \frac{560 \times 560}{560 + 560}$$

$$R_{Th} = 470 + 280 = 750\Omega$$

$$V_{Th} = 10 \times \frac{R_2}{R_1 + R_2} = 10 \times \frac{560}{560 + 560}$$

$$V_{Th} = 5V.$$

For $R_L = 750\Omega$ / $R_L = R_{Th}$ / $R_L = 750\Omega$

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{5}{750 + 750} = 3.33\,mA = 0.00333\,A$$

$$V_L = I_L \times R_L = 0.00333 \times 750 = 2.5$$

$$P_L = V_L \times I_L$$

$$= 2.5 \times 0.00333$$

$$= 8.33\,mW$$

MARKS SPILT UP :-

S.NO	DESCRIPTION	MARKS ALLOTED	MARKS OBTAINED
1)	Experiment	20 Marks	20
2)	Observation	20 Marks	20
3)	Calculation	20 Marks	20
4)	Viva	20 Marks	20
5)	Record	20 Marks	20
		Total	100

100

28/2/26

RESULT:

The maximum power transfer theorem is verified pratically and theoretically.